

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS-601 (GS)

B.E. VI Semester Examination, June 2020

Grading System (GS)

Microprocessor and Interfacing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Calculate delay of following routine:

```
MVI B, 10H
LOOP2: MVI C, FFH
LOOP1: DCR C
JNZ LOOP1
DCR B
JNZ LOOP2
```

2. Assemble the following program, starting at location 2000H

```
START: IN F2H      : Read input switches at port F2H
      CMA          : Set ON switches to logic 1
      ORA A        : Set Z flag if no switch is ON
      JZ START     : Go back and read input port if all switches are off.
```

3. a) Explain Programmable Interrupt Controller in detail.

b) Discuss various modes of DMA transfer.

4. What does it mean "Evaluation of Microprocessor"? Explain various Data Transfer Schemes in detail.

5. Write the initialization instructions of 8259A PIC, to meet the following specifications:

- i) Interrupt type 32
- ii) Edge Triggered, single and ICW4 needed, interval of 8
- iii) Mask IR1 and IR3 interrupts.

6. Why are buffer and latches often required in an 8086 based system? Explain how address, data and control buses of 8086 microprocessor are demultiplexed.

OR

Draw typical 8086 maximum mode configuration and explain the function of signals used in maximum mode.

[2]

7. Given that BX=2500H, SI= 5000H, displacement=1000H and IP=2000H. Determine the effective address for the following addressing modes:
- a) Immediate
 - b) Register using SI
 - c) Direct
 - d) Base
 - e) Index
 - f) Base Index
8. Write short notes on any of the followings.
- a) DMA controller
 - b) 8086 addressing modes
 - c) Flags of 8085
